

**The Access Gap: Final Report on Healthcare Access Risk in Canada
with a Focused Case Study of New Brunswick**

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Executive Summary

Problem Definition and Research Objectives

Canada guarantees universal healthcare in law, but access to that care is unevenly delivered in practice. In the 2022 Canadian Community Health Survey Public Use Microdata File (CCHS PUMF), 26.34% of respondents aged 12 and older meet the operational definition of healthcare access risk: walk-in dependent, without a regular provider, unable to find one, or skipping prescriptions due to cost. New Brunswick is a useful focal jurisdiction because it combines an older-than-average population, two language communities, and one of the longest surgical wait lists in the country. The Access Gap project set out to identify which Canadians face the greatest access risk, where they live, and what individual-level and structural factors explain the variation. The intended audience is the New Brunswick Department of Health together with the Horizon and Vitalité regional health networks; the deliverables are designed to be directly usable by policy and operational decision-makers.

Research Methodology and Research Questions

The project followed the CRISP-DM framework across six deliverables (D1–D5). Three datasets were retained after redundancy testing: the CCHS 2022 PUMF (67,079 respondents, 255 variables), the Open Database of Healthcare Facilities (ODHF v1.1, 7,033 facilities), and the Canadian Institute for Health Information’s 2025 Wait Times release. Three other datasets profiled in D2 were dropped after their contribution proved redundant. The analytical sample, after excluding 22 fully ambiguous records, comprises 67,057 respondents partitioned into 64% training, 16% validation, and 20% sealed test sets through a stratified split with `random_state = 42`. Five research questions guided the work: which factors most strongly predict unmet

healthcare need; how accurately machine learning can predict individual access risk from survey data; how facility distribution across NB health zones relates to access risk; whether proximity to medical education programs correlates with access outcomes; and which New Brunswick populations should be prioritized for investment.

Answering the Research Questions

Household food security emerged as the strongest individual-level predictor in every analysis.

Chi-square independence testing with Cramér's V returned $V = 0.17$ (the largest categorical effect size in the dataset). Point-biserial correlation returned $r = +0.17$. The multivariate logistic regression returned an adjusted odds ratio of 1.44 per category (95% CI 1.40–1.48, $p < .001$), after controlling for age, sex, income, education, immigration status, self-rated health, chronic conditions, and province. A respondent in the most severe food-insecurity category therefore carries approximately 2.95 times the access-risk odds of a fully food-secure respondent.

Self-rated physical health was the second strongest individual-level predictor ($OR = 1.23$, $p < .001$), and household income was protective ($OR = 0.88$ per quintile, $p < .001$). Two findings ran counter to the team's initial expectations: immigrant status was protective ($OR = 0.83$), consistent with the healthy-immigrant effect, and reporting one or more chronic conditions was also protective ($OR = 0.90$), consistent with the increased provider attachment that comes with ongoing care. Geographically, residents of the Territories carried the highest adjusted odds against the Alberta reference ($OR = 3.20$), followed by Prince Edward Island ($OR = 2.11$) and Quebec ($OR = 1.91$). New Brunswick ranked ninth of eleven jurisdictions ($OR = 1.17$, $p = .021$). A tuned LightGBM classifier achieved a positive-class PR-AUC of 0.43 on the sealed test set; at the deployment operating point (threshold = 0.185), the calibrated model identifies 81% of at-risk respondents at 32% precision—appropriate for population-level screening rather than

individual diagnosis. The most policy-relevant finding emerged at the sub-provincial level: within New Brunswick, the Vitalité Edmundston health zone showed an at-risk rate of 35.0% while the Vitalité Bathurst zone showed 20.3%, a nine-point spread inside a single regional health authority.

Implications and Recommendations

Three prioritized recommendations follow from the analysis. First, co-locate primary-care attachment in food-bank and community-meal settings, where the highest-yield at-risk population already concentrates. Second, replace per-capita funding allocation across NB health zones with a zone-weighted formula tied to observed at-risk rates, with Vitalité Edmundston as the priority. Third, expand surgical throughput at Vitalité regional hospitals where wait times are worst. Together these address the dominant individual-level signal, the dominant geographic signal, and the dominant structural signal identified by the analysis.

Discussion

Findings and Insights

The single most important finding of the project is that household food security is a stronger predictor of healthcare access risk than any other measured individual characteristic, including income, education, age, or self-rated health. This held across three independent analytical methods: chi-square / Cramér's V ($V = 0.17$), point-biserial correlation ($r = +0.17$), and multivariate logistic regression (adjusted OR = 1.44 per category, $p < .001$). The effect survives adjustment for nine other features, meaning food insecurity is not simply a proxy for poverty. The result reframes access risk: instead of asking who cannot afford care, the data suggest the

more operationally useful question is who is already disconnected from the stable infrastructure—housing, income, food—that makes primary-care attachment possible.

The second substantive finding is the nine-percentage-point spread in at-risk rates between two zones administered by the same regional health authority. Vitalité Edmundston (Zone 4) recorded 35.0% at-risk, the highest of any New Brunswick zone, while Vitalité Bathurst (Zone 5/6) recorded 20.3%, the lowest. Because both zones operate under one administrative network and share many policy levers, the gap cannot be attributed to management quality. The remaining explanations—local facility density, geographic distance to care, and language-of-service availability—are structural and locality-specific. This finding overturns the framing carried in the project proposal, which described New Brunswick as a province hit harder than most by access shortfalls. After adjusting for individual characteristics, New Brunswick's odds ratio of 1.17 places it ninth of eleven jurisdictions; the operative NB story is intra-provincial, not provincial-average.

Two findings ran against the team's initial expectations and are reported transparently rather than suppressed. Immigrant status was protective ($OR = 0.83$, $p < .001$) after adjustment, consistent with the healthy-immigrant effect documented in Canadian population health literature (Vang et al., 2017) and with the relatively heavy use of community health centres by recent arrivals.

Reporting one or more chronic conditions was also protective ($OR = 0.90$, $p = .003$), which is counter-intuitive until reframed: respondents with chronic illness have more frequent contact with the healthcare system and are therefore more likely to be attached to a regular provider.

Both findings are statistically significant and remain in the model because removing them would distort the estimates on the predictors of primary interest.

Comparison with Initial Hypotheses and Benchmarks

The project began with three operating hypotheses, each of which was modified or refuted by the analysis. The first hypothesis was that the positive class would represent approximately 12% of respondents. Empirical reality is 26.34%—more than double. The discrepancy originated in the proposal’s mental model of unmet need (people without any provider) versus the operational definition used here, which also includes walk-in dependence as an indicator of unstable access. The downstream consequence was substantial: imbalance handling shifted from SMOTE-class techniques to `scale_pos_weight` in LightGBM, and the evaluation framing shifted from severe-imbalance thresholds ($\text{recall} \geq 0.95$) to moderate-imbalance thresholds ($\text{recall} \geq 0.80$). The second hypothesis was that New Brunswick’s provincial average would substantially exceed the national rate. It does not. The 25.5% NB at-risk rate is essentially equal to the 26.3% national rate, and the multivariate adjustment further compresses the difference. The third hypothesis was that proximity to medical education programs would correlate with access outcomes at the health-region level. This question could not be tested at the intended granularity within the PUMF; the geographic resolution required is available only through the CCHS Master file at a Statistics Canada Research Data Centre.

Benchmark comparison is most informative at the provincial level. New Brunswick’s median hip-replacement wait of 256 days in 2022 is the second longest in Canada, well above the 185-day national median reported by CIHI. NB’s knee-replacement wait of 288 days also exceeds the national median of 225 days. Its cataract wait of 72 days, by contrast, is shorter than the 92-day national median. The pattern is consistent with capacity constraints in joint-replacement surgery rather than a uniform supply gap, and aligns with NB’s comparatively low hospital density of 3.4 per 100,000 population versus the national 8.6. The supply-side context confirms a real structural component to the access gap that individual-level demographics alone cannot explain.

Broader Context and Practical Relevance

The findings sit within an established Canadian research tradition on the social determinants of health (Mikkonen & Raphael, 2010; Marmot, 2005). The dominance of food security as an individual-level predictor aligns with PROOF research on household food insecurity in Canada (Tarasuk & Mitchell, 2020), which has independently shown that food-insecure households face elevated downstream healthcare utilization despite reporting greater unmet need. The novel contribution of this project is the demonstration that food insecurity remains the strongest individual-level signal even after adjusting for nine other socioeconomic and health features and ten province dummies. This robustness is what makes the finding operationally useful: targeting food-security touchpoints for outreach is not double-counting income or education effects, it is reaching a population whose access risk is independent of those factors.

The intra-Vitalité finding has direct relevance to NB health authority planning. Resource allocation in regionalized health systems typically follows per-capita formulas; the data here suggest that for primary-care attachment specifically, a risk-weighted allocation would shift modest amounts of capacity from Bathurst to Edmundston and produce measurable improvement in attachment rates. The recommendation does not require new funding—only a reallocation of existing resources within the Vitalité network. This is the kind of operationally actionable finding that justifies the dashboard and report deliverables submitted with this project.

Limitations and Challenges

Challenges Faced

Data-related challenges were the most consequential. CCHS encodes non-response and not-applicable cases as numeric codes (6, 7, 8, 9 for single-digit variables; 96, 97, 98, 99 for derived

variables) rather than as missing values, which meant that a literal-read of the file produced silently corrupted descriptive statistics. The team built a coded-missing replacement step that applied to every CCHS variable except four (SMKDVSTY, HWTDGISW, HWTDGWHO, HWTDGBCC), where the value 6 carries substantive meaning (never smoked, child not applicable). Forty additional variables had to be identified and removed as leakage—including the two target-defining items and thirty DO_* survey-routing flags that encode which optional modules each respondent received. Failure to remove these would have produced an inflated PR-AUC reflecting target leakage rather than predictive signal. The wait-times dataset lacked territory-level rows; a synthetic territories row was added using national-median imputation to preserve the eleven-province structure.

Methodologically, the principal challenge was the moderate class imbalance described above. Three approaches were trialed: SMOTE-NC for mixed categorical/numeric data, class-weighted training, and scale_pos_weight tuning within LightGBM. SMOTE-NC produced unrealistic interpolated values on ordinal CCHS codes; class-weighted training produced equivalent results to scale_pos_weight at higher computational cost. The final pipeline uses scale_pos_weight, tuned in [1.0, 5.0] by Optuna. A second methodological challenge was the decision between native LightGBM categorical handling and one-hot encoding for ordinal features. Three full tuning runs (200 Optuna trials each) confirmed that treating ordinals as integers preserved the ordered-threshold splits the algorithm relies on and produced marginally better PR-AUC than categorical encoding.

Resource constraints shaped two decisions. First, the NBCC compute server required VPN access that proved unreliable from off-campus, so the final Optuna tuning was performed on an eight-core laptop over approximately seventy-five minutes per run. Second, an initial attempt to

use Azure AutoML for benchmarking returned misleading results because its default weighted PR-AUC metric averaged across both classes and was dominated by the negative class. The team reverted to manually tuned LightGBM with positive-class PR-AUC as the explicit objective.

Limitations of the Findings

Generalizability is constrained by the cross-sectional design of CCHS 2022. The associations reported here are correlational, not causal; without longitudinal follow-up the analysis cannot distinguish whether food insecurity causes access risk, whether access risk worsens food insecurity, or whether both reflect a shared upstream factor such as unstable housing. The findings generalize to Canadians aged 12 and older in private dwellings during 2022; they do not generalize to institutional populations, to children under 12, or to time periods substantially before or after the survey wave.

Model accuracy and uncertainty are reported honestly. The sealed test PR-AUC of 0.43 reflects a genuine ceiling on survey-only features and is consistent across three independent tuning runs.

The model is appropriate for population-level screening and outreach prioritization; it is not appropriate for individual diagnosis or for any consequential automated decision-making. At the recommended deployment threshold (0.185), precision is 32%—meaning roughly two of every three flagged individuals would be false positives at the current feature set. Any operational use must include a human review step before action is taken on a flagged record.

External factors limit the supply-side analysis. The ODHF dataset captures facility location but not capacity, hours of operation, or language-of-service. The CIHI wait-times dataset reports medians but not the underlying queue dynamics. Both gaps mean that the engineered provincial supply features are coarser than ideal. The medical education question (RQ4) could not be answered at the intended health-region granularity because the LODE/ODEF facility list was

redundant with ODHF at the provincial level; resolving this requires access to the CCHS Master file through a Statistics Canada Research Data Centre.

Impact of the Limitations and Mitigation Strategies

The most consequential downstream effect of these limitations is on the strength of the recommendations. Because the analysis is associational, the recommendations are framed as targeting hypotheses to be validated through pilot programs rather than as proven causal interventions. Each recommendation includes an implicit evaluation design (attachment rates before vs after a pilot in food-bank settings, at-risk rate change in Edmundston after a zone-weighted reallocation, wait-time change after surgical capacity expansion) that would allow the NB Department of Health to confirm or revise the recommendations as evidence accumulates. Mitigation strategies applied during the project include the strict train/validation/test split with a sealed test set that was never examined during tuning, the use of an explicit anti-leakage variable list updated as new leakage candidates were discovered, the preservation of source variables in a separate parquet file for bivariate analysis (with modeling-side parquets stripped of those variables), and the isotonic calibration of probabilities before threshold tuning. These collectively reduce the risk that reported model performance overstates real-world performance.

Suggestions

Key Insights Recap

Three insights are load-bearing for the recommendations that follow. First, household food security is the dominant individual-level signal, and its effect survives adjustment for nine other features. Second, New Brunswick's access burden is intra-provincial: a nine-point spread inside the Vitalité network is the operative geographic story, not a provincial average that closely tracks

the national rate. Third, supply-side context independently predicts risk: NB's low hospital density (3.4 per 100k vs 8.6 national) and long surgical waits (hip 256 days vs 185 national) confirm a structural component to the access gap that demographic features alone cannot explain.

Prioritized Recommendations

Recommendation 1 (Priority: High; Cost: Low; Cycle: Short). Co-locate primary-care attachment services in food-bank and community-meal settings across New Brunswick. The statistical basis is direct: $V = 0.17$ and adjusted OR = 1.44 per food-security category identify food-insecure households as the highest-yield outreach population. The mechanism is to embed a nurse-practitioner or community health worker at participating food-security organizations and convert the existing visit into a primary-care attachment opportunity. The intervention is low-cost because it reuses infrastructure already serving the target population and low-risk because it does not require new regulatory authority.

Recommendation 2 (Priority: High; Cost: Medium; Cycle: Medium). Replace per-capita funding allocation across NB health zones with a zone-weighted formula tied to observed at-risk rates. The statistical basis is the nine-point Edmundston-vs-Bathurst spread within the Vitalité network. The mechanism is to shift modest amounts of primary-care attachment capacity, transport subsidies, and multilingual reception capability from lower-risk to higher-risk zones. This recommendation does not require new funding; it requires a reallocation formula that the Department of Health and Vitalité can implement within their existing budgets.

Recommendation 3 (Priority: Medium; Cost: High; Cycle: Long). Expand surgical throughput at Vitalité regional hospitals where wait times are highest. The statistical basis is NB's 256-day hip-replacement median (second worst nationally) and a positive point-biserial correlation between provincial hospital density and access risk. The mechanism is operating-room expansion

paired with the zone-weighted allocation in Recommendation 2. This is the highest-cost recommendation and the longest-cycle, but it directly addresses the supply-side gap that engineered ODHF and CIHI features confirm statistically.

Implementation Strategies

Recommendation 1 should be implemented as a six-month pilot in three to five food-bank locations distributed across Horizon and Vitalité networks, with attachment rates measured against a matched comparison group of non-participating food banks. Recommendation 2 requires the development of a transparent zone-weighting formula, ideally jointly by the Department of Health and the two regional health authorities, with the at-risk rate from this project's Power BI dashboard as the weighting input. Recommendation 3 belongs in the next provincial capital plan; it is a multi-year decision that requires capacity modeling beyond the scope of this project.

Potential Benefits

If Recommendation 1 reaches even a modest fraction of food-insecure households without a regular provider, the attachment-rate gain compounds because every newly attached respondent is also less likely to require costly emergency department visits for primary-care-treatable conditions. Recommendation 2 produces an equity benefit at no net cost, reducing the within-province gap between zones. Recommendation 3 addresses both the access metric and the surgical wait metric simultaneously, with downstream benefits in functional health and labour participation. The Power BI dashboard delivered with this project provides the measurement substrate for tracking each intervention's effect over time.

Monitoring and Evaluation

Each recommendation includes a measurement plan. For Recommendation 1, the primary outcome is the attachment rate (percentage of pilot food-bank visitors with a regular provider ninety days after intake) measured against a matched control. For Recommendation 2, the primary outcome is the zone-level at-risk rate measured annually using the next CCHS cycle, with Edmundston as the focal zone. For Recommendation 3, the primary outcome is the CIHI-reported median hip-replacement wait at Vitalité regional hospitals, with secondary outcomes for knee and cataract waits. The Power BI Service report can be extended to surface each of these metrics on a dedicated monitoring page once the underlying data refreshes are available.

Continued Analysis and Future Research

Three follow-up research directions are recommended. First, secure access to the CCHS Master file through a Statistics Canada Research Data Centre to enable health-region-level analysis, which would allow direct testing of Research Question 4 and a more granular version of the within-Vitalité finding. Second, link CCHS to administrative claims data (hospital discharge abstracts, physician billing) to convert the cross-sectional associations reported here into longitudinal estimates of subsequent healthcare utilization. Third, extend the modeling to a multi-cycle CCHS dataset to test whether the food-security signal is stable over time or whether the post-pandemic period materially shifted access patterns. Each of these directions is feasible within an academic or applied research-centre setting and would strengthen the operational case for the recommendations above.

Ethical Disclaimer

Adherence to Ethical Principles

This project was conducted with explicit adherence to data privacy, respect for the populations represented in the data, and fairness in the framing of findings. All datasets used are publicly released by Statistics Canada or the Canadian Institute for Health Information and are anonymized at source. The CCHS Public Use Microdata File contains no direct identifiers; geographic resolution is intentionally limited to the province and (for selected provinces, including New Brunswick) the health-region level. The project did not attempt to re-identify respondents and did not link the public dataset to any external identifying information. The team treated New Brunswick's two regional health authorities as operating systems to be supported, not adversaries to be exposed, and framed every finding accordingly.

Specific Actions Taken

Concrete actions taken during the project to uphold these principles include the following. Geographic findings were reported only at the provincial and (where permitted by PUMF design) health-region level; no attempt was made to drill below the PUMF's intentional resolution limits. The two counter-intuitive findings on immigration and chronic conditions were reported transparently rather than omitted, because suppressing them would have misrepresented the real distribution of risk and risked overgeneralizing the food-security finding to populations where it does not apply. The model was calibrated and threshold-tuned with explicit framing as a population-level screening tool, and the Discussion and Limitations sections explicitly state that the model is unsuitable for individual diagnostic use. The Power BI dashboard delivered with this project carries a visible data-ethics disclaimer on its cover page indicating the analytical and policy-research scope of the findings.

Acknowledged Limitations Despite Best Efforts

Despite these measures, residual ethical considerations remain. The CCHS PUMF, by design, under-represents on-reserve Indigenous populations, certain northern communities, and residents of institutional dwellings. Findings that read as national should be understood as representative of CCHS-eligible Canadians, not of every Canadian. The use of point-biserial correlations and odds ratios can be misread by non-technical audiences as direct predictions about individuals; the team has worked to frame the results in population-level language throughout, but residual misinterpretation risk cannot be fully eliminated. Finally, any recommendation that targets a specific population for outreach carries a small risk of stigma; the food-bank intervention in Recommendation 1 should be implemented with explicit attention to respectful framing and voluntary participation.

Commitment to Continuous Improvement

The team is committed to continuous improvement in the ethical conduct of analytical work. This commitment includes openness to feedback from the NB Department of Health, Horizon, Vitalité, and the academic supervisor on the framing of the findings; willingness to revise the public-facing materials (dashboard, report, presentation) if any framing is shown to produce unintended harm; and willingness to extend the methodology to subsequent CCHS cycles as they are released, so that the recommendations can be re-tested against more recent data rather than fossilized at the 2022 wave.

References

Canadian Institute for Health Information. (2025). Wait times for priority procedures in Canada, 2025: Data tables. CIHI.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.

Marmot, M. (2005). Social determinants of health inequalities. *The Lancet*, 365(9464), 1099–1104. [https://doi.org/10.1016/S0140-6736\(05\)74234-3](https://doi.org/10.1016/S0140-6736(05)74234-3)

Mikkonen, J., & Raphael, D. (2010). *Social determinants of health: The Canadian facts*. York University School of Health Policy and Management.

Statistics Canada. (2023). *Open Database of Healthcare Facilities (ODHF)*, version 1.1. Statistics Canada.

Statistics Canada. (2024). *Canadian Community Health Survey – Annual Component (CCHS), 2022: Public Use Microdata File*. Statistics Canada.

Tarasuk, V., & Mitchell, A. (2020). *Household food insecurity in Canada, 2017–18. Research to identify policy options to reduce food insecurity (PROOF)*. University of Toronto.

Vang, Z. M., Sigouin, J., Flenon, A., & Gagnon, A. (2017). Are immigrants healthier than native-born Canadians? A systematic review of the healthy immigrant effect in Canada. *Ethnicity & Health*, 22(3), 209–241. <https://doi.org/10.1080/13557858.2016.1246518>